

SHREYAS RAORANE

Philadelphia, PA | +1 (445) 345-0829 | raorane@seas.upenn.edu
shreyas0812.github.io | linkedin.com/in/shreyas-raorane | github.com/Shreyas0812

Education

University of Pennsylvania

M.S.E. in Robotics GPA: 3.80/4.0 (Summa Cum laude)

Philadelphia, PA

Aug 2024 – May 2026

Relevant Coursework: Control and Optimization for Robotics, Machine Perception, Learning in Robotics, F1/10 Autonomous Racing

SRM Institute of Science and Technology

B.Tech. GPA: 3.91/4.0 (First Class with Distinction)

Chennai, India

Jun 2018 – Jun 2022

Skills

Programming	C++, Python, MATLAB, Object-Oriented Design, Design Patterns
Systems	Multi-threading, Real-time Systems, Memory Management, Event-driven Architecture, Linux
Robotics	ROS2, Sensor Integration (LIDAR, IMU, RGB-D), Localization, Multi-Agent Coordination
Motion Planning	RRT*, A*, MPC, DMPC, QP Optimization, Sampling-Based Planning, Collision Avoidance
Tools & Sim	Gazebo, PyBullet, AutoDRIVE, Foxglove, Docker, Git, CI/CD

Work Experience

EvoSort Simulation

Jun – Aug 2025

OPEX Corporation, Time Critical Software Engineering Intern (3 Months)

Moorestown, NJ

- Engineered multi-threaded discrete-event simulation in C++ for warehouse automation, managing 1,000+ parallel material flow events with shared state across 3 concurrent threads
- Developed time-critical scheduling engine completing actuation sequences within 3 milliseconds, modeling real hardware timing for production validation
- Refactored legacy codebase with shared header architecture and a centralized `SimulationHandler`, letting the software team validate against the simulator instead of waiting for limited physical hardware availability

ESAP Migration Project

Jun 2022 – Jun 2024

Brillio Technologies, Software Development Engineer (2 Years)

Bengaluru, India

- Developed and maintained the Angular frontend for Verizon's ESAP Migration Project (10,000+ enterprise users), delivering 10+ production screens over 1.5 years
- Optimized 40+ PostgreSQL queries supporting the platform, reducing response time by 35%
- Resolved 50+ QA-filed JIRA defects before merge, maintaining release stability across a large-scale enterprise deployment

Robotics Projects

F1/10 Autonomous Racing Stack

Feb – Mar 2025

University of Pennsylvania

- Architected full ROS2 autonomy stack integrating LIDAR, IMU, and RGB-D for perception, localization, and planning; bridged the AutoDRIVE Simulator and F1/10 hardware over a shared message interface with live parameter tuning
- Repository: Shreyas0812/roboracer-ws

SBAMP: Sampling-Based Adaptive Motion Planning

Mar – Apr 2025

University of Pennsylvania

- Built adaptive RRT* planner with dynamical-systems attractors for real-time goal tracking; replanned fast enough to track goals where static RRT* could not keep up
- Published at the ICRA 2026 Workshop on Frontiers of Optimization for Robotics
Paper: openreview.net/forum?id=7CiufF99ue
- Repository: Shreyas0812/SBAMP

SwarmSync: Distributed MPC for Quadrotor Swarms

Oct – Dec 2025

University of Pennsylvania

- Implemented DMPC for collision-free trajectory generation across 8+ quadrotors using QP optimization
- Agents reliably reached goals in simulation with dynamic obstacles, degrading only in extremely dense environments
- Repository: Shreyas0812/SwarmSync

Research Experience

Master's Thesis: Scalable Multi-Robot Coordination

Oct 2025 – May 2026

University of Pennsylvania, GRASP Lab Advisors: Prof. Linh Phan, Prof. Rahul Mangharam, Prof. M. Ani Hsieh

LCBA Task Allocation with Decentralized Path Planning

- Designed LCBA (Local Consensus Bundle Algorithm), a decentralized task allocation algorithm extending GCBBA to intermittently disconnected, dynamic communication graphs via component-local consensus and event-driven replanning
- Integrated LCBA with three interchangeable path planners (Cooperative A*, PBS, RHCR); evaluated across six maps from 6 to 200 agents over 432+ runs against CBBA, SGA, and DMCHBA, achieving a 0.91 makespan ratio vs. centralized SGA
- Maintained non-zero task completion in scenarios where CBBA and SGA baselines dropped to zero
- Repository: Shreyas0812/LCBA-Warehouse-System